

ML FROM PRIMITIVES

2026 EDITION

# ML Interview Workbook

A compact, printable workbook for ML interview preparation.

**Use this to prepare for the interview you actually have.** Write down the loop, choose the most important gap, attempt it closed-book, record what happened, and schedule the next attempt. The workbook covers planning, practice, simulations, stories, and the final week.

**Candidate:** \_\_\_\_\_

**Loop date:** \_\_\_\_\_

## How to use it

1. **Start with recent practice.** Record confirmed rounds, time available, and what you can do without notes.
2. **Choose three gaps.** Work on the areas most likely to affect your actual loop.
3. **Attempt, compare, retry.** Keep notes closed until the attempt ends; compare your answer with the criteria; schedule the next attempt.
4. **Call a skill ready carefully.** Look for two successful attempts on different days and one unfamiliar follow-up.

## 1. Loop snapshot

**Target role:** \_\_\_\_\_

**Level:** \_\_\_\_\_

**Primary domain:** \_\_\_\_\_

**Weeks available:** \_\_\_\_\_

**Hours per week:** \_\_\_\_\_

**Interview date:** \_\_\_\_\_

### Confirmed round formats

- ☐ ML breadth / fundamentals
- ☐ Modeling / experimentation
- ☐ LLM systems / inference
- ☐ Behavioral / leadership
- ☐ Coding / implementation

- ☐ ML system design
- ☐ Training / debugging
- ☐ Project deep-dive
- ☐ Research / work sample
- ☐ Other: \_\_\_\_\_

## Your route through the field guide

Start with the rounds the recruiter actually named. Use the links below to choose a first lesson, then use the browser Workbook to save your role, round choices, evidence scores, and retry dates. **Practice loop:** read one visual lesson, close it, attempt without notes, compare with the reference, record the gap, and retry after time has passed.

| Round                   | Start here                                                                  | Method and repeat                                                                                                   |
|-------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| ML breadth              | <a href="#">Bias and variance</a> ; <a href="#">choose a loss</a>           | Explain the mechanism, assumptions, failure, and one alternative in eight minutes. Retry in two days.               |
| ML system design        | <a href="#">multi-team platform</a> ; <a href="#">fixed-budget system</a>   | Scope first, then draw data, model, evaluation, serving, monitoring, and rollback. Retry with a changed constraint. |
| Project deep-dive       | <a href="#">ambitious project</a> ; <a href="#">technical project</a>       | Practice the opening, three decisions, ownership, failure, and result. Repeat weekly with push-back.                |
| Coding / implementation | <a href="#">debug a training loop</a> ; <a href="#">implement attention</a> | State the contract, write a baseline, test edge cases, then explain complexity. Retry in two to three days.         |
| Research depth          | <a href="#">critique a paper</a> ; <a href="#">design an ablation</a>       | State the claim, strongest alternative, test, and evidence that would change your mind. Repeat weekly.              |
| Behavioral / values     | <a href="#">project story</a> ; <a href="#">values pressure</a>             | Use a real decision, conflict or failure, result, and update. Retell from prompts, not a script.                    |

## Role paths

Choose the path that matches your target role, then follow its sequence in the browser Workbook:

|                           |                                                                                                                                                                     |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Applied Scientist         | <a href="#">scope</a> , <a href="#">breadth</a> , <a href="#">design</a> , <a href="#">experimentation</a> , <a href="#">ownership</a> , <a href="#">simulation</a> |
| Machine Learning Engineer | <a href="#">coding</a> , <a href="#">debugging</a> , <a href="#">production</a> , <a href="#">system design</a> , <a href="#">failure stories</a>                   |
| Research Engineer         | <a href="#">implementation</a> , <a href="#">systems</a> , <a href="#">research depth</a> , <a href="#">ablations</a> , <a href="#">critique</a>                    |
| Research Scientist        | <a href="#">paper defense</a> , <a href="#">math</a> , <a href="#">experiments</a> , <a href="#">job talk</a> , <a href="#">collaboration</a>                       |

## 2. Evidence scorecard

Score only recent timed, closed-book practice. **0** means not attempted; **3** means you can give a workable timed answer; **5** means two successful attempts on different days plus an unfamiliar follow-up.

| Area                    | 0                        | 1                        | 2                        | 3                        | 4                        | 5 / note |
|-------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|----------|
| Fundamentals            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | _____    |
| Production / systems    | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | _____    |
| LLM systems             | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | _____    |
| Retrieval / search      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | _____    |
| System design           | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | _____    |
| Experimentation         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | _____    |
| Behavioral / leadership | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | _____    |
| Math / research         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | _____    |
| Coding / implementation | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | _____    |

### 3. First three practice priorities

Choose gaps by importance and evidence. Each plan item should name an action, not just a topic.

01      Gap / round: \_\_\_\_\_  
          Closed-book action: \_\_\_\_\_  
          What success looks like: \_\_\_\_\_  
          Retry date: \_\_\_\_\_

---

02      Gap / round: \_\_\_\_\_  
          Closed-book action: \_\_\_\_\_  
          What success looks like: \_\_\_\_\_  
          Retry date: \_\_\_\_\_

---

03      Gap / round: \_\_\_\_\_  
          Closed-book action: \_\_\_\_\_  
          What success looks like: \_\_\_\_\_  
          Retry date: \_\_\_\_\_

---

### 4. Story bank prompts

For each story, keep the ownership and evidence explicit. **Problem / stakes**

\_\_\_\_\_  
\_\_\_\_\_

My decision

\_\_\_\_\_

Alternatives rejected

\_\_\_\_\_

Evidence / result

\_\_\_\_\_

Failure or surprise

\_\_\_\_\_

What I would change

\_\_\_\_\_

## 5. Weekly operating loop

---

|                          |                                                                                                                              |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------|
| extbfMonday              | Pick the three most important gaps. Name the round and what success looks like.                                              |
| extbfTuesday to Thursday | Attempt closed-book. Use a visual lesson only after the attempt. Record what you could not explain, derive, design, or code. |
| extbfFriday              | Retry one weak item and one unfamiliar transfer. Ask one follow-up that changes an assumption.                               |
| <b>Weekend</b>           | Review the evidence, update the plan, and remove work that is no longer relevant.                                            |

---

**This week's focus**

What changed after an attempt?

Next action

## 6. Readiness check

Do not call a critical gap ready based on confidence alone.

- Two successful closed-book attempts on different days.
- No critical scoring area remains weak.
- One unfamiliar follow-up or changed assumption goes well.
- One later mixed practice session remains workable.
- A peer, mentor, or interviewer has challenged at least one claim when possible.

**What I still need to show:**



## 7. Closed-book attempt and retry log

Use one row per meaningful attempt. Record what you could do without notes and what changed after feedback.

| Date | Round | Prompt or task | Score | Gap | Retry |
|------|-------|----------------|-------|-----|-------|
|      |       |                |       |     |       |
|      |       |                |       |     |       |
|      |       |                |       |     |       |
|      |       |                |       |     |       |
|      |       |                |       |     |       |
|      |       |                |       |     |       |
|      |       |                |       |     |       |
|      |       |                |       |     |       |
|      |       |                |       |     |       |

## Retry protocol

|                    |                                                          |
|--------------------|----------------------------------------------------------|
| Before the attempt | What must I do without notes? _____                      |
| After the attempt  | What could I not do yet? _____                           |
| Next practice      | What is the smallest task that addresses that gap? _____ |
| Evidence           | What should improve next time? _____                     |

## 8. Simulation review

Use this after a role-specific simulation or a full practice round. Separate a smooth answer from evidence that holds when an assumption changes. **Simulation / role:** \_\_\_\_\_ **Date:** \_\_\_\_\_

\_\_\_\_\_

**Rounds included:** \_\_\_\_\_

**Time limit and conditions:** \_\_\_\_\_

\_\_\_\_\_

What held under pressure? \_\_\_\_\_

\_\_\_\_\_

Where did I lose structure? \_\_\_\_\_

\_\_\_\_\_

Which assumption changed the answer? \_\_\_\_\_

\_\_\_\_\_

What evidence was missing? \_\_\_\_\_

\_\_\_\_\_

What will I practice before the next simulation? \_\_\_\_\_

\_\_\_\_\_

---

## Observer / interviewer feedback

**Strongest signal**

\_\_\_\_\_

**Most important improvement**

\_\_\_\_\_

**Follow-up I should rehearse**

\_\_\_\_\_





## 9. Project and story practice

Keep the story focused on decisions, ownership, evidence, failure, and what you changed. Do not memorize a script.

90-second opening                      Problem, ownership, outcome, and the three decisions I will explain.

Decision 1                                  Constraint: \_\_\_\_\_

Choice            and            evidence:  
\_\_\_\_\_

Decision 2                                  Constraint: \_\_\_\_\_

Choice            and            evidence:  
\_\_\_\_\_

Decision 3                                  Constraint: \_\_\_\_\_

Choice            and            evidence:  
\_\_\_\_\_

Ownership                                  What did I personally decide, build, unblock, or change?

Failure / surprise                      What failed, and how did it update the plan?

Impact and guardrail                      What changed for users or the business, and what could have regressed?

---

## Changed-assumption follow-ups

**01. If the data, latency, budget, or label changes:**

---

**02. If the first approach fails:**

---

**03. If a senior partner disagrees:**

---

## 10. Other interview rounds

Use this page for formats outside the guide. Those formats need their own practice plan.

| External format                        | Separate resource or drill | Baseline date |
|----------------------------------------|----------------------------|---------------|
| General algorithms / DSA               | _____                      |               |
| Practical software / OOP / concurrency | _____                      |               |
| SQL / data manipulation                | _____                      |               |
| General distributed systems            | _____                      |               |
| Other:                                 | _____                      |               |

## 11. Final week

The final week is for steady practice on known gaps, not broad new coverage.

- ☐ Recheck each critical round with one closed-book baseline.
- ☐ Retry only the most important weak areas.
- ☐ Run one mixed simulation with an unfamiliar follow-up.
- ☐ Practice the 90-second project opening and three decisions.
- ☐ Confirm time zone, links, format, environment, and permitted tools.
- ☐ Prepare concise questions about scope, success, collaboration, and the first 90 days.
- ☐ Stop adding broad topics; protect sleep and response quality.

**Logistics / questions / risks**



## 12. Review and backup

At the end of each week, update the plan from what you practiced, not from how you felt.

What did I complete?

---

What remains critical?

---

What should I remove or defer?

---

What is next week's one priority?

---

---

### Backup reminder

The browser workbook is local to the device. Export its JSON backup before changing devices or clearing browser data. This printable PDF contains no private plan data.

## A compact operating rule

**Practice without notes. Compare your answer with the criteria. Work on one gap. Retry after time has passed.** Study a topic only when it changes what you can do in the interview.